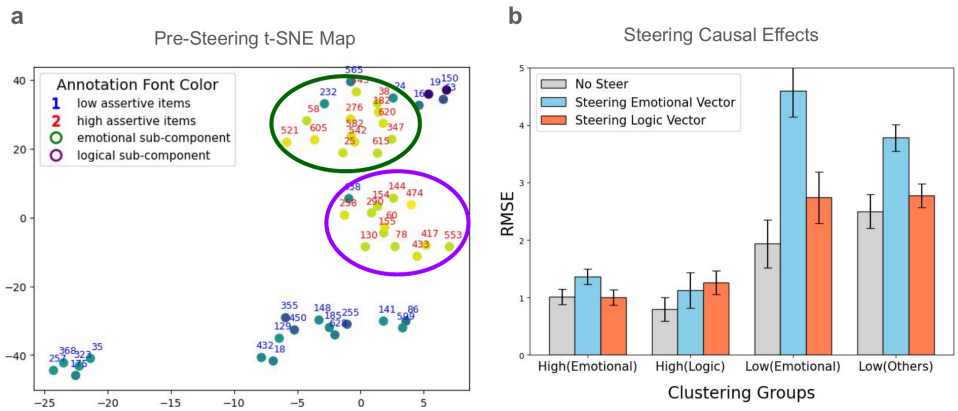


Figure 3: (a) t-SNE visualization of assertiveness clusters, showing low-assertive items (blue indices) and high-assertive items (red indices). Green and purple circles highlight emotional and logical sub-components, respectively (manual labels based on local grouping). Circle fill color indicates standardized assertive score, with lighter shades for higher scores. (b) RMSE changes after removing emotional or logical sub-component vectors, shown separately for high-assertive-logical, high-assertive-emotional, low-assertive-emotional (e.g. index 232, 16, 19), and other low-assertive items (e.g. index 35, 432, 141). Error bars are standard error of the mean (SEM).

With this borderline and layer setting, t-SNE visualizations revealed two distinct clusters within the high-assertive items (n = 25) (Figure 3a). Based on ChatGPT summaries, one cluster was labeled "emotional (big green circle)" and the other "logical (big purple circle)," aligning with the Elaboration Likelihood Model, a long-standing dual persuasion model in Psychology [14]. The theory posits that the logical cluster corresponds to central-route persuasion (evidence, statistics, facts), while the emotional cluster corresponds to peripheral-route persuasion (affective or superficial cues). The logical sub-component partially overlapped with certainty expressions, but which had minor influence on model predictions in these datasets.

Adding extra 25 low-assertive items (blue indices) in the t-SNE map showed partial overlap with the emotional cluster but not the logical one (Figure 3a), suggesting partial independence of the logical sub-component. High-assertive sub-components were spatially cohesive and clearly separating from low-assertive items, reflecting a decision boundary in assertiveness representation space.

Removing the emotional vector broadly degraded performance statistically (i.e. changed beyond SEM error bars), particularly on all low-assertive items and emotionally-relevant high-assertive cluster. In contrast, removing the logical vector primarily affected predictions for logical high-assertive items, indicating a more localized effect (Figure 3b).

## 4 Conclusion and limitations

We present an empirical method for localizing and interpreting LLM activations linked to assertiveness along a continuous scale. Results show that high-assertive representations split into two orthogonal sub-components—emotional and logical persuasion—each exerting distinct effects when used as steering vectors. Yet, this study has several limitations. The model analyzed is relatively small (1B parameters), leaving scaling effects uncertain, and only one architecture is examined, so cross-model generality remains untested. The dataset size is modest (n = 645), which may obscure finer-grained cues. Moreover, cluster interpretation relies on manual inspection, suggesting that future work should develop more automated approaches. Despite these limitations, this study provides the first visualization of correlated feature decomposition along a continuous assertiveness scale. It offers new methodological insights for analyzing LLM internal representations and contributes to understanding the roots of overconfident behaviors in LLMs.

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